

Genetic mutations underlying isoniazid-resistant *Mycobacterium tuberculosis* in Khyber Pakhtunkhwa, Pakistan

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ABSTRACT

Tuberculosis, caused by *Mycobacterium tuberculosis*, is a major public health issue in Pakistan. Isoniazid is a first-line pro-drug that requires activation through an enzyme called catalase peroxidase, but is subject to widespread resistance, driven by mutations in *katG* and *inhA* genes and other loci with compensatory effects (e.g., *ahpC*). Here, we used whole genome sequencing data from 51 *M. tuberculosis* isolates collected from Khyber Pakhtunkhwa province (years 2016–2019; all isoniazid phenotypically resistant) to investigate the genetic diversity of mutations in isoniazid candidate genes. The most common mutations underlying resistance were *katG* S315T (37/51), *fabG1* -15C>T (13/51; *inhA* promoter), and *inhA* -154G>A (7/51). Other less common mutations ($n < 5$) were also identified in *katG* (R128Q, V1A, W505*, A109T, D311G) and candidate compensatory genes *ahpC* (-54C>T, -51G>A) and *oxyS* (M249T). Using DynaMut2 software, the mutants exhibited various degrees of stability and flexibility on protein structures, with some *katG* mutations leading to a decrease in KatG protein flexibility. Overall, the characterisation of circulating isoniazid resistant-linked mutations will assist in drug resistant TB management and control activities in a highly endemic area of Pakistan.

1. Introduction

Tuberculosis (TB) disease, caused by *Mycobacterium tuberculosis* bacteria, affects ~10 million people annually [1]. Despite reductions in TB incidence, increasing drug resistance, particularly to first-line isoniazid, is making infection control difficult. New diagnostic and molecular tools are needed to infer drug resistance and characterise genetic diversity across different geographical regions. The detection of *M. tuberculosis* resistant to first-line rifampicin (RR-TB) and isoniazid, together called multidrug resistance (MDR), is especially crucial in Pakistan, as it ranked eighth globally in terms of burden [1,2]. Previous

work has considered the diversity of genetic mutations associated with rifampicin resistance, namely in the *rpoB* gene, where S450L and V483G variants are in high frequency [2]. Unlike rifampicin, isoniazid is a pro-drug that inhibits the formation of the mycobacterial cell wall. Isoniazid must be activated by KatG, a bacterial catalase-peroxidase enzyme in *M. tuberculosis* [3]. The mechanism of action of isoniazid involves inhibition of the synthesis of mycolic acids by acting through the NADH-dependent enoyl-acyl carrier protein (ACP)-reductase, encoded by *inhA* [3]. Isoniazid resistance therefore involves *katG* and *inhA* genes, but also other loci (*oxyS-ahpC*, *ahpD* and *ahpE*) [3]. For example, it is known that *inhA*, *ahpC* and *ahpE* genes can mitigate the

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initial fitness cost caused by *katG* mutations, and *ahpC-oxvR* mutations may be directly linked to resistance [4]. The overexpression of *ahpC* and *inhA* may not affect the conferred resistance to isoniazid [4].

Whole-genome sequencing is revolutionising the study of *M. tuberculosis* genetic diversity and drug resistance, including through characterising known and novel mutations in candidate resistance genes [5], identifying new loci through GWAS [6,7] and machine learning methods [8,9], and understanding the transmission of resistance through phylogenetic techniques [10]. Establishing the function or impact of putative resistance variants on protein structures can be facilitated through *in silico* modelling (e.g. using DynaMut2 software [11]), which predict the stabilizing and destabilizing effects of mutations on thermodynamic properties of biomolecules [11]. The combining of protein structural modelling and genomic diversity can provide new insights into resistance mechanisms and potentially inform the development of new anti-TB drugs [12]. In this study, we evaluated the genetic diversity of mutations in *katG* and other isoniazid candidate genes, and assessed their impact on protein stability. We focus on isolates that have undergone whole genome sequencing, and been sourced from the Khyber Pakhtunkhwa province in Pakistan. This region has only just re-established TB control activities after years of armed conflicts [2,13], and is known to have high isoniazid resistance prevalence [13]. By characterising the circulating genetic variation and their potential functional effects, we hope to provide a baseline for future whole genome sequencing initiatives and inform the design of diagnostic tools.

2. Materials and methods

Khyber Pakhtunkhwa has an estimated 270 TB cases per 100,000 population. Between years 2016 and 2019, there were ~8000 *M. tuberculosis* samples collected per year at the Provincial TB Reference Laboratory, Hayatabad Medical Complex Peshawar. Retreatment cases make up ~16% of TB observed. A convenience sample of fifty-one isolates (enriched for MDR) spanning years 2016–2019 were selected from the Provincial TB Reference Laboratory biobank. All samples were subject to phenotypic drug susceptibility testing (DST), performed through the automated BACTEC MGIT 960 system (BD Diagnostic Systems, NJ, USA) (14), confirming all 51 isolates as isoniazid resistant. The study was approved by the Institutional Ethics Committee of Kohat University of Science and Technology, Kohat, and Provincial Tuberculosis Reference Laboratory, Khyber Pakhtunkhwa (Ref. no.: PTP/PTRL-408/19). Genomic DNA for the fifty-one isolates was extracted by using the acetyl trimethylammonium bromide (CTAB) procedure (see [2] for details). The extracted DNA was subjected to whole-genome sequencing at the London School of Hygiene & Tropical Medicine (LSHTM) through The Applied Genomics Centre using the Illumina MiSeq platform with a 151bp paired end protocol. All sequence accession numbers are listed (S1 Table).

All raw reads were filtered for quality using the FastQC tool (v0.11.8), and aligned to the H37Rv genome (Accession No. AL123456.3) using BWA-mem (v0.6) software (<http://bio-bwa.sourceforge.net>). Single nucleotide polymorphisms (SNPs) and insertions and deletions (indels) were called using the GATK tool "Haplotype Caller" (gatk.broadinstitute.org), and the resulting variants were annotated for function (e.g., non-synonymous mutations, frame shifts). Variants with less than 10-fold coverage were marked as missing. Further, the functional effects of mutations on KatG, AhpD, and AhpE protein structures were computed using the online DynaMut2 server (<http://biosig.unimelb.edu.au/dynamut2/>), leading to measures of the effect on protein stability (changes in vibrational entropy, $\Delta\Delta G$: kcal/mol) and flexibility ($\Delta\Delta S_{vib}$: kcal/mol⁻¹K⁻¹). A multiple sequence alignment was constructed using SNPs found by mapping to the reference genome. Across each sample, the alignments were concatenated into a single alignment using in-house scripts. Phylogenetic reconstruction was based on a maximum-likelihood approach implemented using iqtree software (v1.6.12, -m GTR+G+ASC) (www.iqtree.org). The TB-Profiler was used

to predict *M. tuberculosis* genotypic drug resistance and lineages (including mixed-lineage infections) [5,14]. The obtained results were compared with a mutational database consisting of ~34k *M. tuberculosis* covering all major sub-lineages and drug resistance types [14].

3. Results

Of the 51 *M. tuberculosis* isolates, the majority were sourced from female patients (n = 29, 56.9%) and re-treatment cases (n = 30, 58.8%) (S2 Table). Six isolates were phenotypically assessed as resistant to isoniazid, but did not have any known resistance or candidate mutations that could explain resistance. A total of 6128 SNPs were identified across the 51 isolates, leading to a phylogenetic tree with the expected lineage-based clustering, where the largest clade represented lineage 3 (30, 58.8%) (Fig. 1). Mixed-lineage infections were detected by TB-Profiler in four isolates (S1 Table). The underlying mutations responsible for resistance in the isolates were *katG* S315T (37/51), *fabG1* -15C>T (13/51; *inhA* promoter), and *inhA* -154G>A (7/51). Other less common mutations (n < 5) were also identified in *katG* (e.g., R128Q, V1A, W505*, A109T, D311G) and in candidate compensatory *ahpC* (-54C>T, -51G>A) and *oxyS* (M249T) genes. Additionally, although the importance of indels for drug resistance has been previously highlighted [15], no resistance-associated indels were observed in the 51 isolates. Several mutations appear to be unique to this study (*katG* R128Q, *oxyS* M249T, *ahpD* N128T and *ahpE* P135Q), being absent in a global 34k isolate dataset (Table 1). Mutations co-occur in the same isolates, with *katG* S315T co-occurring with *ahpD* (N128T), *ahpE* (P135Q) and *oxyS* (D142 N), potentially contributing compensatory roles (S3 Table). Some of the mutations (e.g., *inhA* -154G>A and *oxyS* M249T) appear in different clades in the phylogenetic tree, providing evidence of convergent evolution, and supporting their association with a resistance or compensatory function (Fig. 1). Whilst other mutations appear to be present in isolates clustering closely on the phylogenetic tree, which could be indicative of transmission events (e.g., *ahpE* P135Q in samples with identical SNP patterns), or phylogenetically related (e.g., *katG* R463L) (Fig. 1).

The effects of the point mutations on protein structure dynamics were predicted using DynaMut2 software (S1 Figure; Table 1). The N128T mutation, present in the loop region of the AhpD protein, exhibited increased flexibility in the surrounding helices and loops, which may render it as a weak target or loss of ligand-protein interaction. Total energy estimates suggest that the N128T mutation has a stabilizing effect on the AhpD protein. Similarly, mutations R128Q and D311G in KatG led to increasing molecule flexibility, but with destabilizing effects. In contrast, the established resistance *katG* S315T mutation, as well as *katG* A109T and *ahpE* (P135Q), had predictions of decreasing molecule flexibility and destabilizing effects on their respective proteins.

4. Discussion

The increasing prevalence of drug resistant *M. tuberculosis* is making TB treatment and control difficult. The growing routine use of whole-genome sequencing as a diagnostic will not only inform clinical decision making and facilitate timely treatment, but will highlight the genetic diversity of circulating *M. tuberculosis*. In this study, we focused on isoniazid resistance, which is a major issue in Pakistan. We aimed to characterise mutations in a sample of 51 isolates enriched for isoniazid resistance, with all being phenotypic resistant. Most isoniazid resistance in the 51 isolates could be attributed to *katG* mutations, particularly S315T (37/51), which is the most common mutation globally (~75%) and in Pakistan (~90%) [13]. We also identified rare mutations such as *katG* R128Q, *oxyS* M249T, *ahpD* N128T and *ahpE* P135Q, which were not observed in our larger global dataset of 34k isolates. The rarity of *katG* R128Q mutation, paired with the fact it co-occurs with a compensatory mutation, indicated that it might come with a high fitness

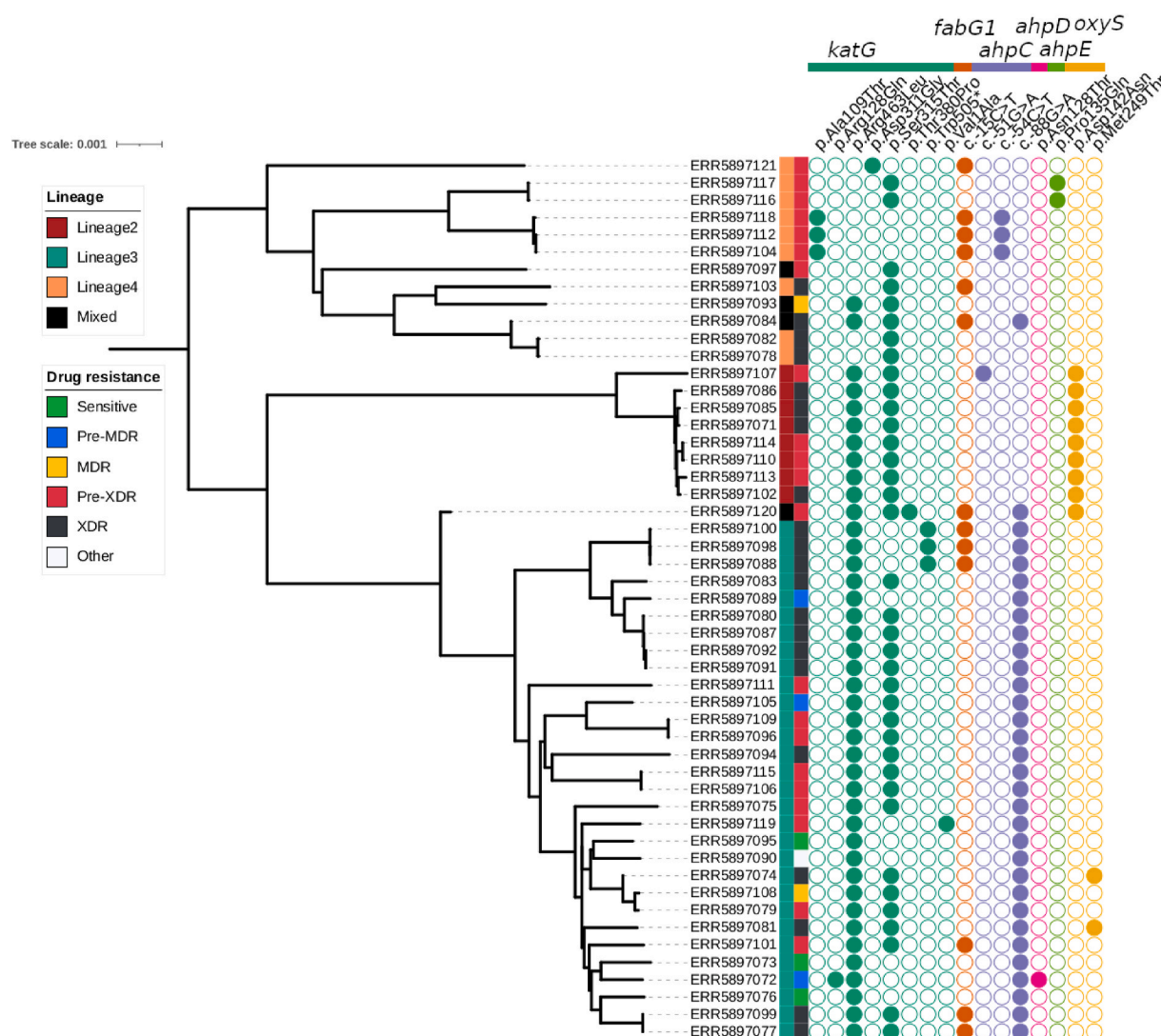


Fig. 1. Phylogenetic tree for the study samples ($n = 51$) with mutations in drug resistance candidate genes.

Solid circles indicate the presence of a mutation. Pre-MDR = resistance to isoniazid or rifampicin; MDR = resistance to isoniazid and rifampicin; Pre-XDR resistance = MDR + resistance to a fluoroquinolone or a second-line injectable; XDR = MDR + resistance to a fluoroquinolone and a second-line injectable. Drug resistance types are consistent with the WHO definitions at the time of sample collection.

cost, which prevents its widespread acquisition in drug resistant isolates. Both *katG* R128Q and T380P have not previously been associated with resistance but occur in the same codon as known resistance mutations and could explain the discrepancy between the resistant phenotype and the *in-silico* (genotypic) predictions. The roles of these mutations will need to be further characterised to ascertain if they cause resistance or have a compensatory effect. The known compensatory mutation *ahpC* –54C>T co-occurs with the rare *katG* A109T, suggesting that compensatory mutations may occur at higher frequency in isolates with rare *katG* resistance mutations. The evidence for a prominent compensatory role of *ahpD*, *oxyS* and *ahpE* loci in isoniazid resistance is less strong. Our study identified only single *ahpE* and *ahpD* mutations, all potentially unreported and found in the presence of *katG* mutations. Additionally, several mutations were found to bear the signatures of lineage specific mutations, including *oxyS* D142 N and the known lineage-mutation *katG* R463L [16]. Other mutations such as *oxyS* M249T display evidence of convergent evolution, hinting to a potential selection pressure associated with increased fitness. In general, potential *ahpE*, *ahpD* and *oxyS* compensatory mutations and mechanisms need to be validated experimentally in fitness studies, or through large-scale studies involving phylogenetic analysis of longitudinally collected strains.

By modelling the predicted effects of mutations on protein function using DynaMut2 tools, the results suggest potentially damaging effects caused by *katG* (S315T, A109T) and *ahpE* (P135Q) mutations, where *katG* S315T is an established resistance variant. In general, combining GWAS and population genomic methods with predicted protein function effects can reveal putative resistance markers and loci, as well as high-light targets for anti-TB drugs that may be robust to resistance [17]. In addition, hetero-resistance was observed in isolates involved in mixed-lineage infections. This complexity could contribute towards poor outcomes, as previously hypothesised [18]. Six isolates with phenotypic resistance did not have known or candidate resistance mutations. Considering the well characterised resistance genes for isoniazid, the phenotype was probably incorrectly assigned as resistant. This highlights the potential value of whole-genome sequencing to provide accurate resistance calls in clinical settings. Indeed, a recent multicentre cohort study found widespread discrepancies between local phenotypic tests and sequencing-based genotypic resistance [19]. Overall, our work has provided a baseline characterisation of potential isoniazid resistance-associated mutations in *M. tuberculosis* circulating in Khyber Pakhtunkhwa province, and supports future whole-genome sequencing based investigations to inform clinical and infection control in a high TB

Table 1

Mutations found with their effects protein structure stability and flexibility.

Gene	Genome position	Mutation	Count ^a	Polar	Hydrophilic	Energy kcal/mol ^b	Flexibility kcal.mol ⁻¹ .K ^{-1c}	Global N ^d	Global isoniazid resistance % ^e
<i>katG</i>	2154724	R463L	41	P-NP	HP-HB	-0.191	0.060	15987	40.4
<i>katG</i>	2155168	S315T	37	P-P	HP-HP	1.012	-0.207	7203	99.1
<i>katG</i>	2155787	A109T	3	NP-P	HB-HP	0.531	-0.557	14	81.8
<i>katG</i>	2154597	W505Stop	3	-	-	-	-	2	100
<i>katG</i>	2156110	V1A	1	NP-NP	HB-HB	#	#	12	87.5
<i>katG</i>	2155729	R128Q	1	P-P	HP-HP	-1.003	0.432	0	-
<i>katG</i>	2155180	D311G	1	P-NP	HP-HB	-0.751	0.471	1	-
<i>katG</i>	2154974	T380P	1	P-NP	HP-HB	#	#	5	100
<i>fabG1</i>	1673425	-15C>T	13	-	-	-	-	2023	93.1
<i>inhA</i>	1674048	-154G>A	7	-	-	-	-	336	89.0
<i>oxyS</i>	141623	D142 N	9	P-P	HP-HP	#	#	7480	60.2
<i>oxyS</i>	141945	M249T	2	NP-P	HB-HP	#	#	0	-
<i>ahpC</i>	2726139	-54C>T	3	-	-	-	-	45	93.8
<i>ahpC</i>	2726142	-51G>A	1	-	-	-	-	48	97.0
<i>ahpD</i>	2727188	N128T	1	P-P	HP-HP	0.378	0.381	0	-
<i>ahpE</i>	2510773	P135Q	2	NP-P	HB-HP	0.456	-0.111	0	-

P polar, NP non-polar, HP hydrophilic, HB hydrophobic.

Bolded mutations are mutations that have not previously been associated with drug resistance (using the TB-Profiler *tbdh* resistance database).

Residue at this position has not been modelled.

^a Number of isolates in that have the mutation.^b Total energy ($\Delta\Delta G$: kcal/mol), where +ve values are stabilizing and -ve values are destabilizing.^c $\Delta\Delta S_{vib}$ ENCoM (kcal/mol⁻¹K⁻¹), where negative values reflect decreasing molecule flexibility.^d Total number of samples which have the mutation, based on a global 34k isolate dataset [14].^e % of samples from the global dataset with an isoniazid drug susceptibility test that were resistant.

burden setting.

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Ethical approval and consent to participate

The present investigation was approved by the Institutional Ethics Committee of Kohat University of Science and Technology, Kohat, and Provincial Tuberculosis Reference Laboratory, Khyber Pakhtunkhwa.

Consent for publication

Not applicable.

Author contributions

TGC and TAK conceived and directed the project. ASK, MTK, SAI, MQ, NM, SAH, BK, SC and TAK contributed data. ASK, JEP, MTK and GN performed bioinformatic and statistical analyses under the supervision of SC, TGC and TAK. ASK, JEP, MTK, SC, TGC and TAK interpreted results. ASK, JEP, MTK, TGC and TAK wrote the first draft of the manuscript with inputs from other authors. All authors commented and edited on various versions of the draft manuscript and approved the final version. TGC and TAK compiled the final manuscript.

Declaration of competing interest

All the authors have no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tube.2022.102286>.

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